ME 448 - Lab Project Report:

Ice Rink Design & Analysis

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1 - Introduction

1.1 - Project background

Dr. Pascual has decided to attempt to recreate the ice rink that Poly Royal used in the 1960s. In order for Dr. Pascual to reach his goal this endeavor must be proved to be practical from both an engineering and a finifnatal prospective. To this end, with the provided modeling, data, and component options we have set out to validate the idea of reinstating the Cal Poly ice rink.

The process of designing the thermal systems relevant portions of this ice rink then validating them is a multifaceted task that will be summarized but not fully described herein. Briefly, what we will go over are the heat load calculations where we assess the thermal dynamics of the proposed system and the condition analysis performed in 2D to assess the steady state thermal behavior of the system.

1.2 - EES issues

Prior to this class both Terak and Marcus had little to no EES experience. While we work well together, value each other's contributions, and have no regrets in partner selection, we think it would have been valuable to require each group to have at least 1 person with significant EES experience. The vast majority (over 95%) of the project was spent debugging EES issues, repeating work EES deleted, and generally having our focus pulled away from the actual thermal systems aspect of the project by EES issues. One important thing that this interfered with was our ability to use the EES codes as a design tool, as it took all the time we had to make them work at all.

Several times throughout this duration of this project work we did was lost due to EES issues. An estimated 10 times we lost a negligible amount of work due to VPN crashes. Due to this (and some other unrelated previous experiences) our EES work was compulsively saved very often while working on it. Unfortunately, 1 key time we lost a ton of work when EES decided to save our code as a blank file. This was not a VPN problem as that was carefully checked the day of due to constant experiences with that specific problem. We are still unsure why this happened. Going forward past this point we saved the file as a .ees file and separately (due to last of trust in EES) saved the text itself of the code. While this worked it didn't work well as it did not save the unit system, variable information, formatting and other things like that.

Overall this lost work, VPN issues, and steep learning curve EES possesses were extremely disheartening and led to a loss of motivation, feelings of depressed frustration and a significant lowering on the learning experience we both had. As this was such a proportionally large part of our project experience we elect to include it here, in the report.

2 - Data & Discussion

2.1 - Part 1 of deliverables (head loads)

Table 1: Heat load EES results and part 1 of the deliverables

Parameter:	**8 hour model:**	**24 hour model:**	**Units:**
Power required to freeze ice	38501	12843	BTU/hr
Power required to cool the ice (to 24 °F)	1063	354.5	BTU/hr
Power required to cool the water (from 52 to 32 °F)	5411	1804	BTU/hr
Heat gain by natural convention on the ice surface	3941	3941	BTU/hr
Heat gain by radiation exchange on the ice surface	20661	20661	BTU/hr
Total startup power	69577	39594	BTU/hr
Total startup power	5.798	3.2995	Tons
Total steady state power	24602	24602	BTU/hr
Total steady state power	2.050167	2.050167	Tons

The results in the table indicate a clear distinction between the energy requirements of the 8-hour and 24-hour models. The 8-hour model demands significantly more power to freeze the ice, cool it to 24 °F, and lower the water temperature, with a total startup power of 69,577 BTU/hr compared to 39,594 BTU/hr for the 24-hour model.

This is due to the faster cooling requirements of the 8-hour design. However, steady-state power remains identical for both models at 24,602 BTU/hr, as the heat gains from natural convection and radiation are constant. The findings suggest that while the 8-hour model achieves quicker freezing, the 24-hour model is more energy-efficient during startup. Further analysis is needed to optimize these parameters.

2.2 - Finite difference

2.2.1 - Input parameters for FEA

- Material information:
  - Ice
    - Thermal conductivity **[0.108 BTU/h*in*^{O}F]**
    - Specific heat **[0.4952 BTU/lb*^{O}F]**
    - Density (not needed for results, but needed for program to run) **[0.0329 lb/in^3]**
  - Insulation
    - Thermal conductivity **[0.00173 BTU/h*in*^{O}F]**
    - Specific heat **[10]** (rough estimate)
    - Density (not needed for results, but needed for program to run) **[1]**
  - Concrete
    - Thermal conductivity **[0.0674 BTU/h*in*^{O}F]**
    - Specific heat **[0.239 BTU/lb*^{O}F]**
    - Density (not needed for results, but needed for program to run) **[1]**
- Boundary conditions:
    - Temperature of ice **[24 ^{O}F]**
    - Temperature of ground **[62 ^{O}F]**
    - Temperature of outside of tubing **[20.66 ^{O}F]**
- Loads:
    - q_startup per unit area (in square inches) differing for each of the 2 startup times. This is dependent on spacing as the spacing will define the reliant area. Results from table 1 were used for this.

2.2.2 - FEA setup

The inputted geometry consists of 3 parts. The ice, insulation, and concrete. This was defined in the FEA software ABAQUS with a 2D drawing of a rectangle with a half circle cut in it to represent the tubing. This geometry is shown below in Figure 1. After extruding a unit depth the overall shape was partitioned into the different material sections, thus modeling the overall setup as a single composite part. This was done because when the various materials were imported as separate parts ABAQUS would not model the heat flux interaction between the adjoining materials. A single part partitioned into segments was the solution found after many hours of trial and error.

Once the geometry was fully defined, materials were created with the specific properties listed above and assigned to the appropriate sections. Then a steady state heat transfer step was defined. Next the mesh was defined with an approximate global spacing of 0.075 inches and the part was meshed dependently (on the part not the instance) as shown in Figure 1.

Temperature boundary conditions were applied to the top, bottom, and tube surfaces. A heat flux per unit area load was applied to the top surface as well representing the startup loading. This load was applied 2 times, one from each start up time requirement. The results of this FEA study can be seen in Figure 2-5 below.

2.2.3 - FEA results

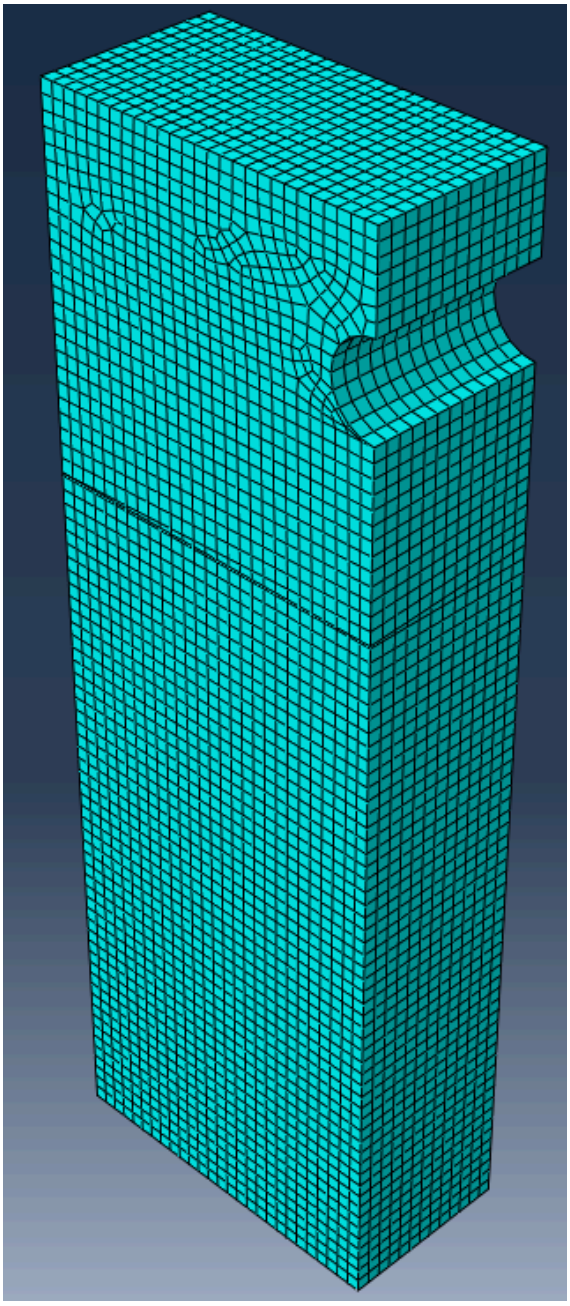


Figure 1: FEA mesh for both time models

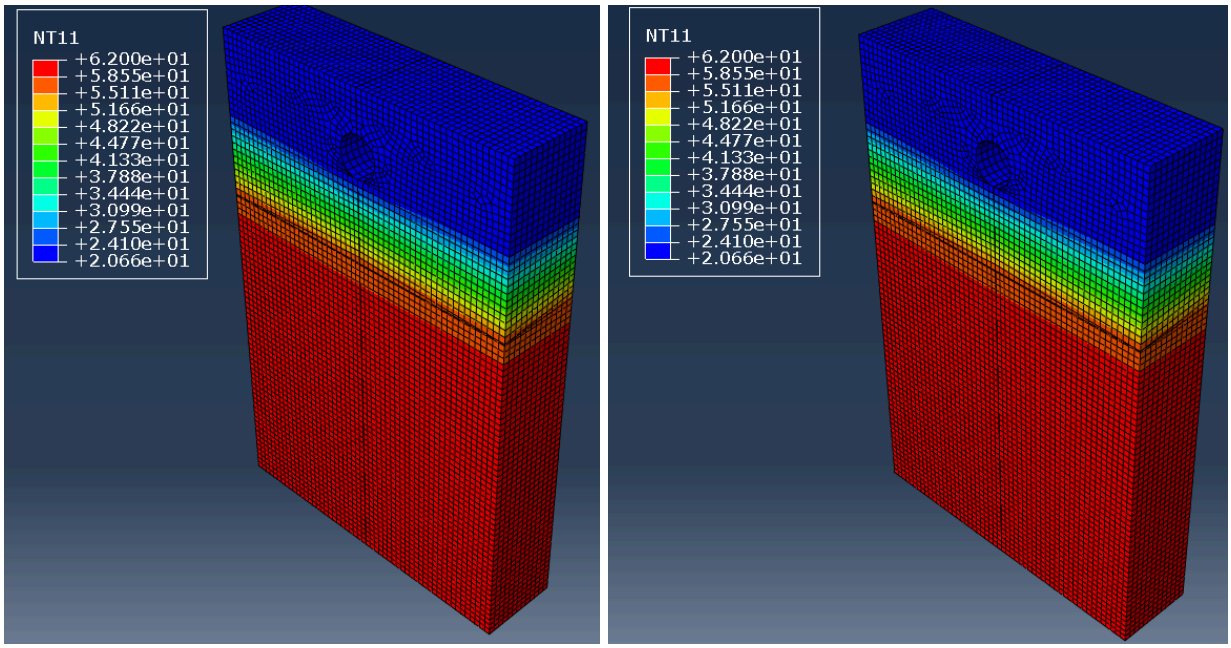


Figure 2: FEA output nodal temperature distribution (8 hour model left, 24 hour model right)

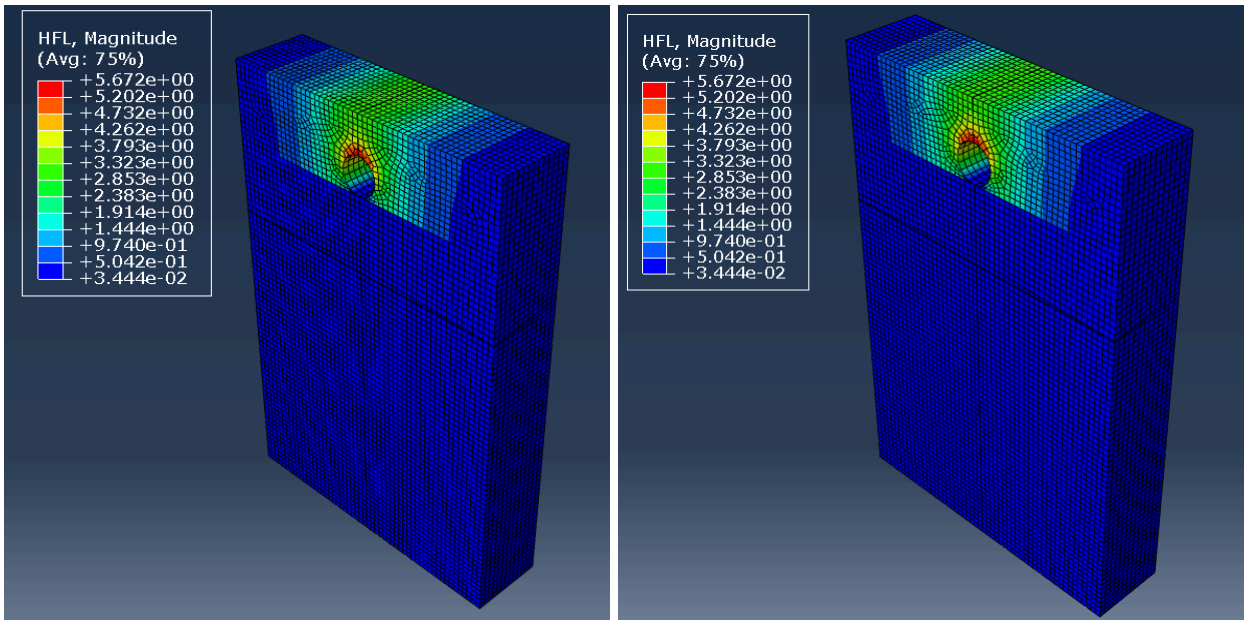


Figure 3: FEA output heat flux magnitude distribution (8 hour model left, 24 hour model right)

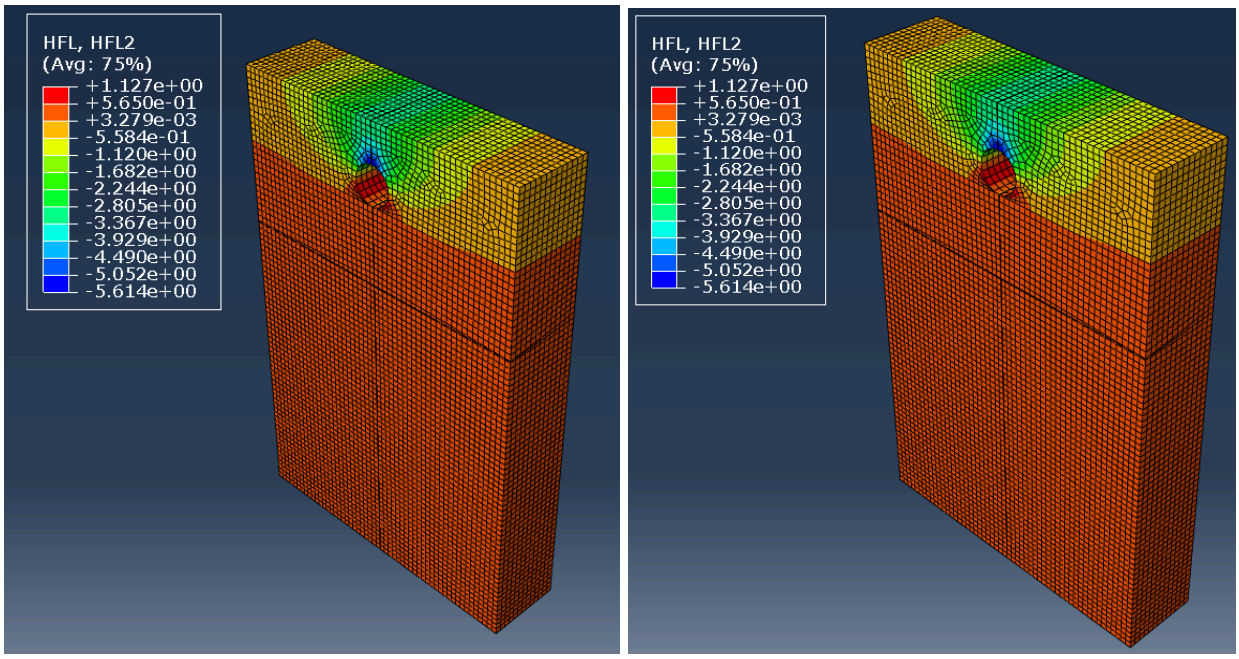


Figure 4: FEA output heat flux in the vertical direction distribution (8 hour model left, 24 hour model right)

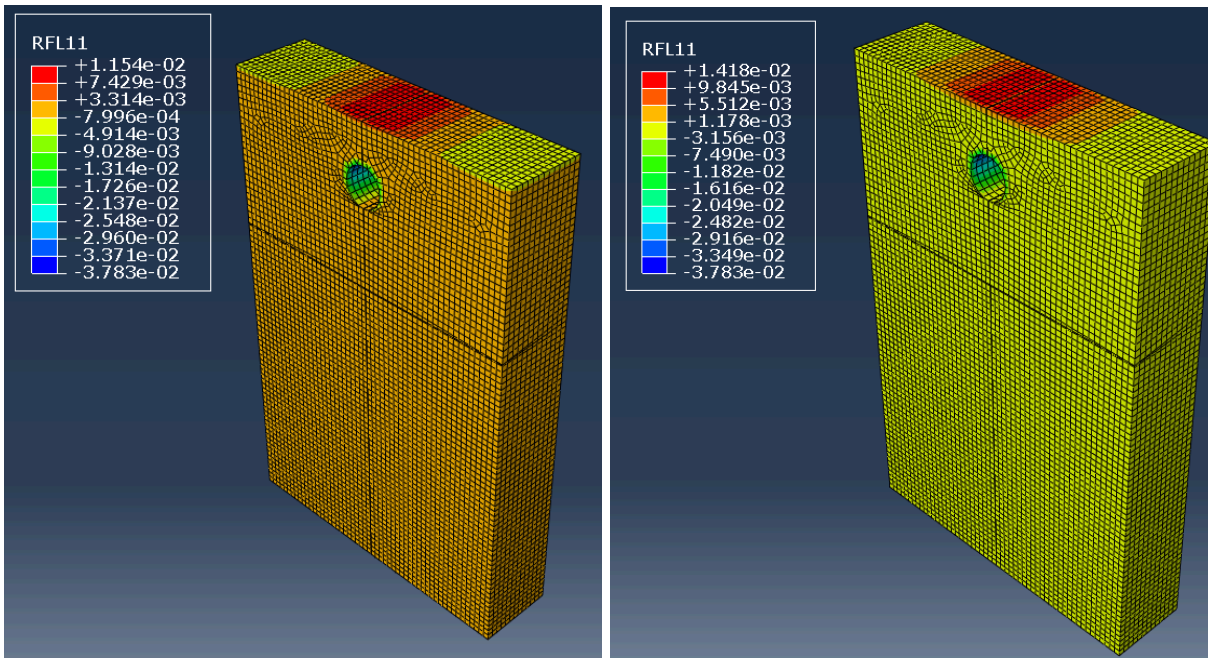


Figure 5: FEA output reaction heat flux magnitude distribution (8 hour model left, 24 hour model right)

Table 2: FEA results and part 2 of the deliverables

Parameter:	**8 hour model:**	**24 hour model:**	**Units:**
Finite difference equations NEEDED to solve the model:	None were needed (FEA)	None were needed (FEA)	NA
Nodal Spacing used in the model:	0.075 (approx)	0.075 (approx)	inches
Surface heat flux of the top of the ice:	2.04192	2.04192	BTU/hr*ft^2
Total heat loss to the ground:	0.477216	0.793728	BTU/hr*ft^2
Final tube OD:	0.5	0.5	in
Final tube spacing:	3	3	in
Tube surface temperature:	20.66	20.66	°F
Thermal resistance of the ice:	4.897351512	4.897351512	hr*ft^2/BTU
Thermal resistance of the insulation & concrete:	15.67152484	15.67152484	hr*ft^2/BTU

The final tube spacing was chosen because it meets the needed requirements. While true constrained optimization has not been done in this project the values and system behavior as well as cost considerations seem to come out well with the chosen spacing value.

2.3 - Heat Transfer in Pipes

Table 3: Quantitative results regarding heat transfer in the piping

Parameter:	**8 hour model:**	**24 hour model:**	**Units:**
Final length of each tube loop	105	105	ft
The flow rate through each loop	5.098	5.098	lbm/hr
The flow rate through each loop	0.00053411	0.00053411	gpm
The total heat transfer rate to the refrigeration system	24602	24602	BTU/hr
The required loop inlet temperature	-34.05	-34.05	°F
The temperature drop though each loop	14.05	14.05	°F

2.4 - System Pressure & Pumping

Pump Selection:8-hour Scenario: Grundfos CR 15-7. 24-hour Scenario: Grundfos CR 15-8. Both pumps operate at approximately 70% efficiency, consume around 2.403 kW, and feature stainless steel construction compatible with CaCl2 brine. They meet the required flow rates and head pressures while operating within the specified temperature ranges.

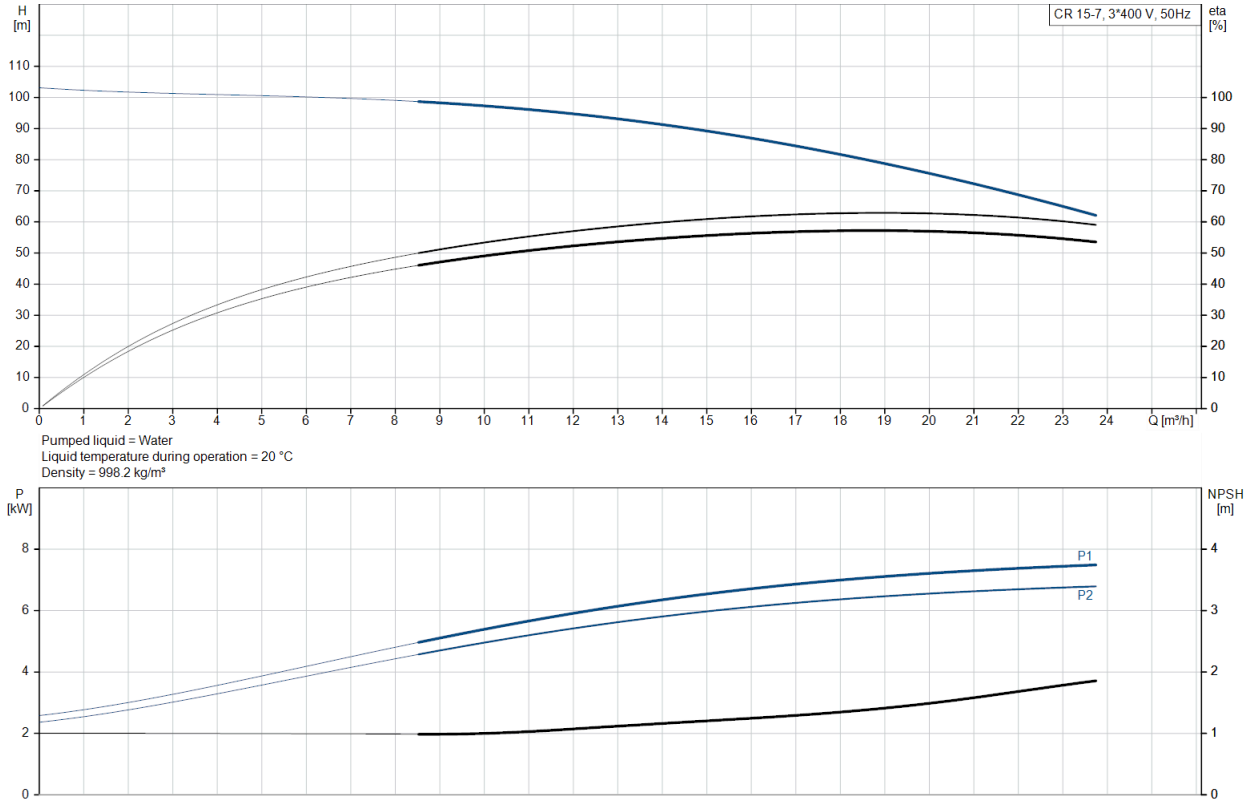


Figure 6: Pump curves for Grundfos CR15-7

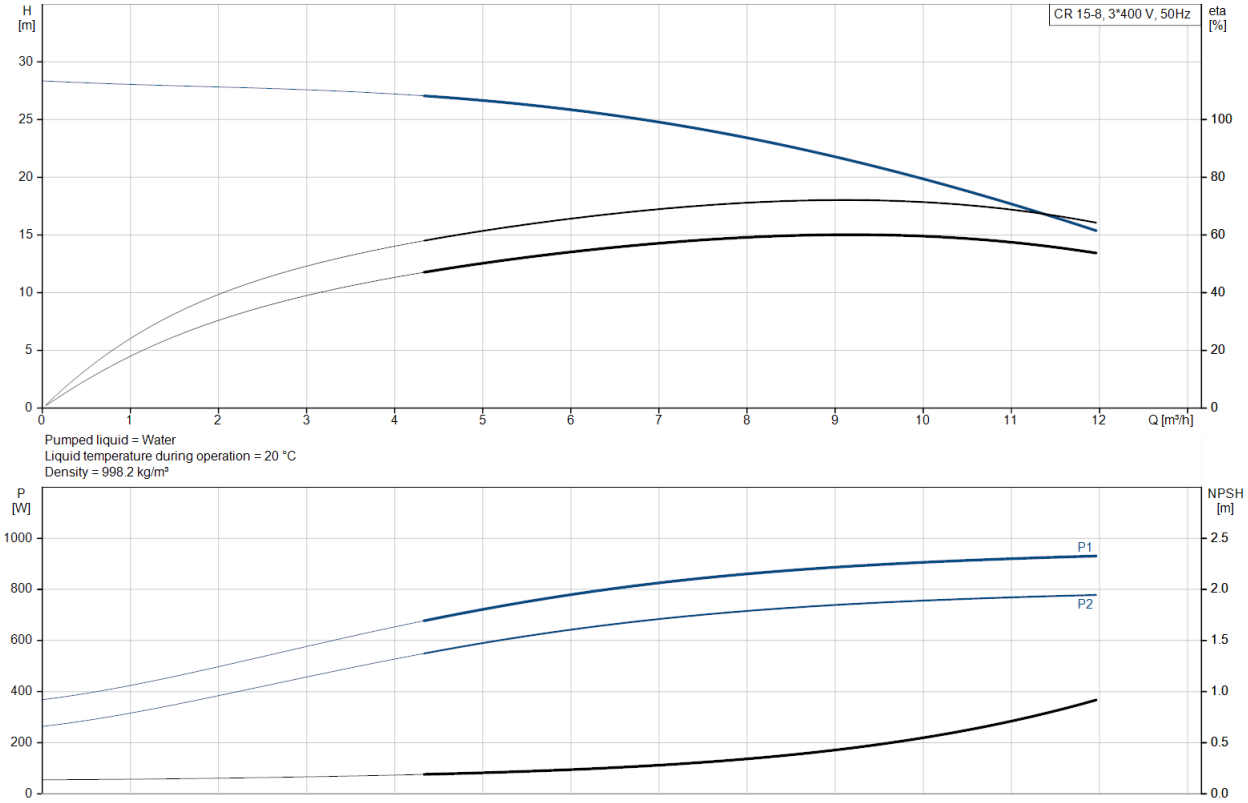


Figure 7: Pump curves for Grundfos CR15-8

Table 4: Quantitative pressure and pump information

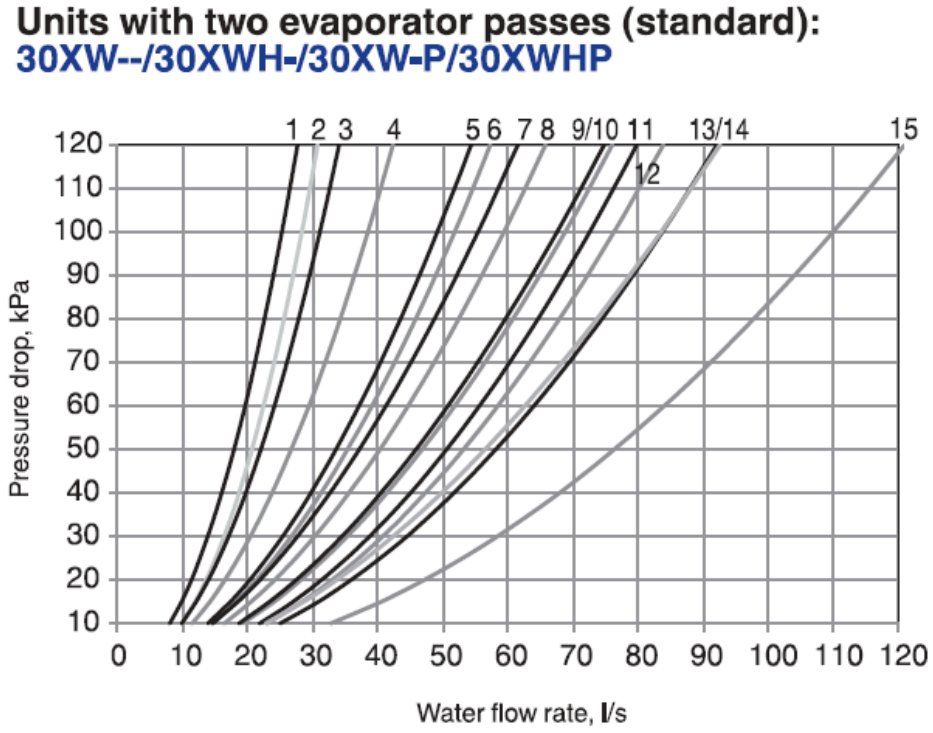
Parameter:	**8 hour model:**	**24 hour model:**	**Units:**
Total system pressure drop	267.5928618	267.5928618	ft
Total system pressure drop	115.979167	115.979167	psi
The pumps total mechanical power	1464	1464	ft*lbf/s
The pumps total mechanical power	2.662	2.662	HP
The pump efficiency	0.7	0.7	NA

2.5 - Chiller & Expansion Tank Selection

Chiller Selection:

• 8-hour Scenario: The Carrier 30XW-H Screw Chiller offers a high cooling capacity with a COP of 3.5, ensuring rapid cooling and energy efficiency. Its stainless steel heat exchanger resists CaCl2 brine corrosion.

• 24-hour Scenario: The Carrier AquaSnap 30MPW Water-Cooled Chiller provides a sufficient safety margin with a COP of 3.8, optimizing energy consumption during extended operations. Its stainless steel plate heat exchanger ensures durability against brine corrosion.

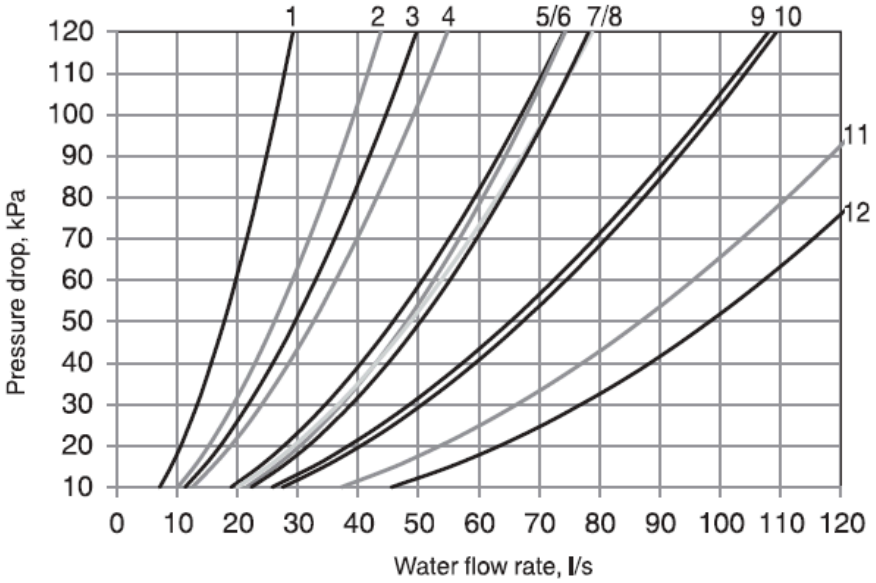


Legend

1. 254
2. 304
3. 354
4. 402, 452, 552, 602
5. 512, 562
6. 652, 702, 802
7. 852
8. 1002, 1052
9. 1154
10. 712, 812, 862
11. 1012,1162
12. 1252, 1352, 1452, 1552
13. 1314, 1464
14. 1652, 1702
15. 1612, 1762

Figure 8: Evaporator and condenser pressure drop curves for Carrier 30XW-H

Units with two condenser passes (standard):
30XW--/30XWH-/30XW-P/30XWHP



Legend

1. 254, 304, 354
2. 402, 452, 552, 602
3. 512, 562
4. 652, 702, 802
5. 712, 812, 862
6. 852
7. 1154
8. 1002, 1052
9. 1012,1162
10. 1252, 1352, 1452, 1552, 1314, 1464
11. 1652, 1702
12. 1612, 1762

Figure 9: Evaporator and condenser pressure drop curves for Carrier AquaSnap 30MPW

Evaporator Water Pressure Drop Curve, Unit Size 033 Only — SI

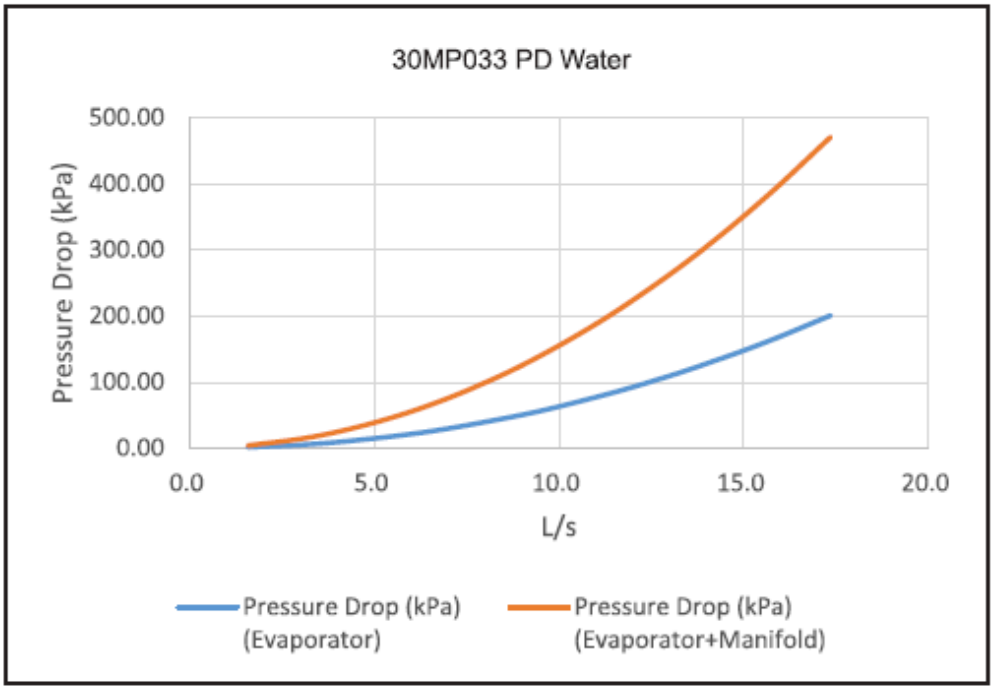


Figure 10: Evaporator and condenser pressure drop curves for aqua snap 30 mpb

Expansion Tank Calculations

Given: $P_{\text{fill,abs}} = 29.7$ psi, $P_{\text{max,abs}} = 54.7$ psi.

8-hour Scenario

$$\Delta V = V_{\text{system}} \times \left(\frac{1}{\rho_{\text{min}}} - \frac{1}{\rho_{\text{max}}} \right) \times \rho_{\text{max}}$$

$$\Delta V_{8hr} = 66.84\,\text{ft}^3 \times \left(\frac{1}{1188} - \frac{1}{1192} \right) \times 1192 \approx 0.225\,\text{ft}^3$$

$$V_{\text{tank,min}} = \frac{0.225\,\text{ft}^3}{\left(\frac{54.7}{29.7} - 1\right)} \approx 0.321\,\text{ft}^3$$

$$V_{\text{tank,selected}} = 0.321\,\text{ft}^3 \times 1.2 \approx 0.385\,\text{ft}^3$$

24-hour Scenario

$$\Delta V = V_{\text{system}} \times \left(\frac{1}{\rho_{\text{min}}} - \frac{1}{\rho_{\text{max}}} \right) \times \rho_{\text{max}}$$

$$\Delta V_{24hr} = 66.84\,\text{ft}^3 \times \left(\frac{1}{1187} - \frac{1}{1190} \right) \times 1190 \approx 0.169\,\text{ft}^3$$

$$V_{\text{tank,min}} = \frac{0.169\,\text{ft}^3}{\left(\frac{54.7}{29.7} - 1\right)} \approx 0.241\,\text{ft}^3$$

$$V_{\text{tank,selected}} = 0.241\,\text{ft}^3 \times 1.2 \approx 0.289\,\text{ft}^3$$

Selected Expansion Tanks

- **8-hour Scenario**: Amtrol THERM-X-TROL ST-5 ($\approx$0.4 ft^3)
- **24-hour Scenario**: Wessels FXA-100 ($\approx$0.3 ft^3)

Both tanks handle the required expansion, pressures up to 125 psig, and are compatible with $CaCl_2$ brine.

Table 5: Quantitative chiller selection information

Parameter:	**8 hour model:**	**24 hour model:**	**Units:**
Overall conductance (UA)	1318.8	1318.8	BTU/hr*ft
Refrigerant temperature	-34.05	-34.05	°F

3 - Price Estimation

We were tasked with calculating the capital cost of the system including overhead and labor for both designs using an online subscription based software called RSmeans. Unfortunately, this is not possible. This is due to the fact that RSmeans provides no information regarding uncertainty in the prevented values. Evidence of this can be found in appendix E below. Without a known uncertainty in the numbers our final cost calculation taking this into account is as follows:

Total cost for the 8 hour model: somewhere between 0 and infinity.

Total cost for the 24 hour model: somewhere between 0 and infinity.

It may be reasonable to accept that the error in the presented values compared to what it would actually cost for the system could be small when it comes to the price of a particular pipe or valve (although it is still possible that these are extremely wrong as well). It is not reasonable to assume that this error would be small when it comes to something like labor cost. As such, as engineers we do not feel that it is appropriate to use this method.

We instead recommend pricing with real vendors and sources of labor local to SLO directly instead of relying on an opaque black box estimation tool like RSmeans. Not only will this provide an exact price but it will also simultaneously start the process of actually getting the project bult. In short this RSmeans based loose estimation is an unnecessary and potentially harmful (if decisions are made based on these uncertain estimates) approach to cost analysis.

In addition to the documented lack of uncertainty information here is a short and non-comprehensive list of some of the important factors related to price that are not able to be taken into account. Justifications for each claimed issue are provided as sub bullets.

- When it would be built
    - The prince of components, labor and everything else depends not only on location but also time
- How quickly it would need to be built
    - Allowing for a slower and less time constrained construction process can reduce costs
- Whether it would need to be disassembled and reassembled
    - Would the labor involved be a relevant cost to consider
    - Perhaps it could be models as a maintenance cost if it were set up/taken down at regular intervals
- Weather Cal Poly has personnel resources (ether staff or students) who would be able to contribute to the bullying like this ME 448 class contributed to the analysis and validation
    - Students are a good source of cheap labor as we are generally desperate for money and hard workers
    - Staff (like Dr. Pascual) could work on this project for free (free in the snake that it would not affect the project budget as they are paid in separately)

    - Note: We are both willing to assist in installing those connectors for ~$40/connector.
- Whether the workers will be unionized
    - Union work cost more generally
- Werther the job will be done as a government bid or in a more economy efficient way
    - This affects the cost significantly
- Whether there will be educational or bulk discounts for some of the components
    - Many companies offer significant discounts for projects like this
- Additional engineering work needed
    - Will it be safe/strong structurally not just in the thermal systems sense
    - Will lawyers get involved in the process needing engineering drawings to be expensively produced and expensively checked by the government.
- Whether anything will be fabricated in house
    - For example some of the connectors that are so experience could be 3D printed at the shop by students instead of bought

It is due to a combination of these reasons (as well as those we did not feel were worthy of inclusion here) and the uncertain uncertainty in the RSmeans data that leads us to be unable to specify a narrower range for the cost then what is given above.

4 - System

4.1 - Requested Brief System Control discussion

The brine system herein disturbed will be controlled automatically. This will require additional sensors and actuators in the system as well as a system control methodology. Single port pressure sensors hooked up to T connectors could be implemented in the brine piping after the pump, before the chiller as well as at the expansion tank. Electrically actuated adjustments to control the compressor and pump could be implemented. Additional control valves could be installed in the main loop near the expansion tank as well. As overshooting would be problematic but time constants are less important with this application a PI control scheme could be used to take in data form the sensors and directly control the actuators. In order to do this a transfer function would need to be calculated analytically so as to ensure the control scheme is stable and to properly select the gain values associated with the proportional and integral portions of the feedback loop respectively.

4.2 - System diagram

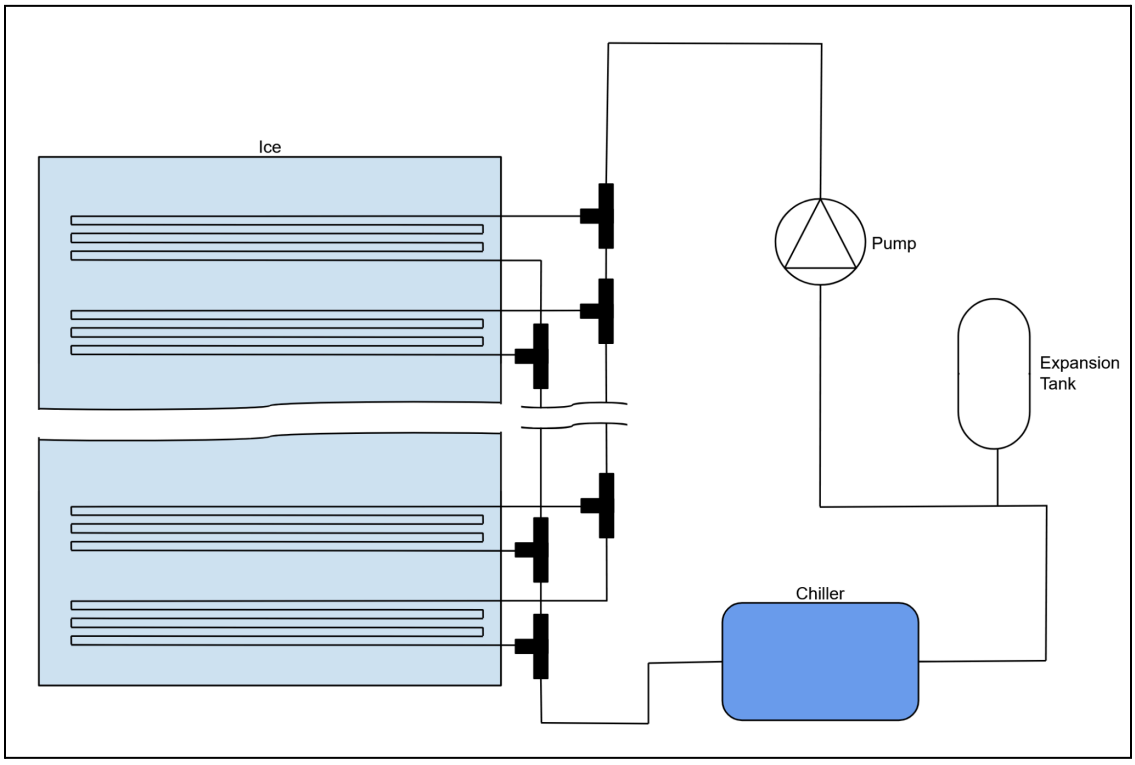


Figure 6: System diagram

5 - Conclusions

5.1 - Project conclusion

The design and analysis of the portable ice rink system have been thoroughly developed, incorporating heat transfer, fluid dynamic, and mechanical considerations. The project's primary objective was to balance efficiency, functionality, and feasibility while meeting the requirements of two operating scenarios: 8 hour and 24 hour startup times. To solve this problem many techniques were used including iterative analysis, EES modeling, and analytical modeling. Key findings and design parameters were established.

Considering heat transfer the total startup and steady-state power requirements for both scenarios were calculated, highlighting the energy trade-offs between rapid freezing (8 hour) and energy efficiency (24 hour). The 8 hour model demands higher startup power due to accelerated freezing, whereas the 24 hour model achieves greater energy savings during startup. Boht perform the same at steady state. Thermal resistances of the ice and insulation, surface heat flux, and overall conductance values were derived to guide system design and refrigeration efficiency.

Considering the fluid dynamics and piping system other conclusions were reached. Piping dimensions, flow rates, and pressure drops were partly optimized for efficient brine circulation and efficient chilling. The selected pumps for each scenario ‘Grundfos CR 15-7’ for the 8 hour model and ‘Grundfos CR 15-8’ for the 24 hour model, allow for reliable and energy-efficient operation, creating the required flow rates and meeting the pressure head requirements. Calculations validated the pressure differences between pipes, tubes, and components, ensuring adequate pumping power and efficiency.

Considering chiller and expansion tank selection brings up further topics. Appropriate chillers were selected to accommodate the cooling loads for each scenario’s unique needs. The water-cooled chiller models were chosen based on thermal capacity, refrigerant temperature, and other considerations, ensuring efficient heat removal from the brine and thus efficient cooling of the ice/water. Expansion tanks were selected to account for brine density changes, ensuring the fluidic aspects of the system maintained integrity and stability during operation.

The system's overall design and component selections demonstrate consistency between models and ensure a balance between performance, cost, and energy efficiency. The methodology included assumptions to simplify the problems when possible.The 8 hour model allows for rapid ice freezing and quicker operational readiness but at the cost of higher power requirements. Alternatively, the 24 hour model comparatively prioritizes lower energy consumption during startup, aligning with sustainable and cost-effective long-term use.

Generally, we are satisfied with the results of our design and analysis thus far. The initial calculations and concepts have provided valuable insights into the performance and feasibility of the system. However, we recognize that further analysis is necessary to refine and optimize

key parameters to approach a truly ideal/optimal solution. While our design seems feasible without a systematic and deeper exploration of the design space we can not be sure it is optimal. The tools herein used have been somewhat effective, utilizing advanced software with more robust and transparent capabilities could improve the implementation of our models, allow for the performing of more complex simulations, and ensure accuracy/exhaustivity in our optimization efforts.

If this project were continued in the future, as seems likely, we have some recommendations. One is to conduct some physical testing to validate the theoretical and analytical (with lots of assumptions) results, especially for the chiller and pump. A deeper exploration of the control systems side of the design could also significantly improve the overall quality and completeness of the design. For example a more adaptive control system could have a significant effect on the efficiency of the system. Similarly, the overall structure should be analyzed. Electrical systems should also be considered as the scale of this project and the power needed to cool the water and freeze the ice is above what can be simply plugged into a standard wall outlet without thought. Design considerations for setup, removal, and storage efficiency provide additional avenues to explore. The insulation materials is another parameter that could be improved to improve the overall system performance.

6 - Appendices

Appendix A - First EES Script

```
"-----------------------------------------------------------------------------------------------------------------------------------"
"ME-448 Heat Load Calculations Lab 1 - Updated"
"-----------------------------------------------------------------------------------------------------------------------------------"
"By Marcus and Terak"
"-----------------------------------------------------------------------------------------------------------------------------------"
"Given Variables"
"-----------------------------------------------------------------------------------------------------------------------------------"
L = 30 [ft]
W = 15 [ft]
A = L * W
Height = 8 [ft]
thickness = 1 [in]*convert(in,ft)
T_ice = 24 [F]
T_ice_R = (24 + 459.67) [R]
T_initial_R = (52 + 459.67) [R]
T_final_R = (32 + 459.67) [R]
T_initial = 52 [F]
T_final = 32 [F]
T_air_R = (80 + 459.76) [R]
T_air = 80 [F]
T_film = 1/2 * (T_ice + T_air)
T_tent = 90 [F]
T_tent_R = T_tent + 459.67 [R]
T_film2 = 1/2 * (T_tent + T_air)
T_sky = 55 [F]
T_sky_R = (55 + 459.67) [R]
rho_ice = density(Ice_inc, T=T_ice)
m_ice = rho_ice * A * thickness
t = 24 [hr]
g = 32.2 [ft/s^2] * convert(ft/s^2, ft/hr^2)
P_atm = 14.7 [psia]
B = 1 / converttemp(F, R, T_film)
B2 = 1 / converttemp(F, R, T_film)
nu = kinematicviscosity(Air, T=T_film, P=P_atm)
nu2 = kinematicviscosity(Air, T=T_film2, P=P_atm)
```

```
alpha = thermaldiffusivity(Air, T=T_film, P=P_atm)
alpha2 = thermaldiffusivity(Air, T=T_film2, P=P_atm)
k = conductivity(Air, T=T_film)
k2 = conductivity(Air, T=T_film2)
cp_water = cp(Water, T=T_final, P=P_atm)
Gs = 291.6 [Btu/hr*ft^2]
sigma = 5.670374419e-8 [W/m^2*K^4]  "Stefan-Boltzmann constant converted to appropriate units if necessary"
epsilon_tent = 0.87  "Assuming this for the tent material"
epsilon_ice = 0.95   "Assume emissivity for ice"
epsilon = 0.87
L_c = A / (2 * (L + W))
R_a_L_c = g * B * (T_air - T_ice) * (L_c^3) * (1 / (nu * alpha))
Nu_L_c = 0.52 * R_a_L_c^(1/5)
h = (Nu_L_c * k) / L_c
R_a_L_c_2 = g * B2 * (T_tent - T_air) * (L_c^3) * (1 / (nu2 * alpha2))
Nu_L_c_2 = 0.15 * R_a_L_c_2^(1/3)
h_tent = (Nu_L_c_2 * k2) / L_c
"-------------------------------------------------------------------------------------------------------------------------------"
"1a) Power required to freeze the ice in Btu/hr for both design conditions (8 hour startup and 24 hour startup)"
"-------------------------------------------------------------------------------------------------------------------------------"
HF = enthalpy_fusion(Water)
q_freeze = (m_ice * HF) / t
"-------------------------------------------------------------------------------------------------------------------------------"
"1b) Power Required to cool ice to 24F in Btu/hr"
"-------------------------------------------------------------------------------------------------------------------------------"
cp_ice = specheat(Ice, T=T_ice, P=P_atm)
q_ice_cool = (m_ice * cp_ice * (T_final_R - T_ice_R)) / t
"-------------------------------------------------------------------------------------------------------------------------------"
"1c) Power Required to cool water from 52F to 32F"
"-------------------------------------------------------------------------------------------------------------------------------"
q_cool_water = (m_ice * cp_water * (T_initial_R - T_final_R)) / t
"-------------------------------------------------------------------------------------------------------------------------------"
"1d) Heat Gain by natural convection on ice surface in Btu/hr"
"-------------------------------------------------------------------------------------------------------------------------------"
```

```
q_conv = h * A * (T_air_R - T_ice_R)
q_tent = h_tent * A * (T_tent_R - T_air_R)
"-------------------------------------------------------------------------------------------------------------------------------
----------"
"1e) Heat gain by radiation on the ice surface in Btu/hr."
"-------------------------------------------------------------------------------------------------------------------------------
----------"
alpha_abs = 0.39 [-]
q_rad_exch = epsilon * sigma * A * (T_tent_R^4 - T_air_R^4)
q_rad_ice = sigma * (T_tent_R^4 - T_ice_R^4) / (((1 - epsilon_tent) / (epsilon_tent * A)) + (1 /
(F_tent_ice * A)) + ((1 - epsilon_ice) / (epsilon * A)))
q_rad_surr = sigma * (T_tent_R^4 - T_sky_R^4) / (((1 - epsilon_tent) / (epsilon_tent * A)) + (1 /
(F_tent_surr * A)))
q_surr_ice = sigma * (T_sky_R^4 - T_ice_R^4) / (((1 - epsilon_ice) / (epsilon_ice * A)) + (1 /
(F_ice_surr * A)))
q_rad = q_rad_ice + q_rad_surr
q_net = q_rad_ice + q_rad_surr
"Assumed flow fractions"
F_tent_ice = 0.5  "Assumed fraction of heat transfer to ice via tent"
F_tent_surr = 1 - F_tent_ice
F_ice_surr = 1 - F_tent_ice
"-------------------------------------------------------------------------------------------------------------------------------
----------"
"1f) Total startup power required in Btu/hr and tons for both designs."
"-------------------------------------------------------------------------------------------------------------------------------
----------"
q_startup = q_freeze + q_ice_cool + q_cool_water + q_conv + q_rad
"-------------------------------------------------------------------------------------------------------------------------------
----------"
"1g) Total steady state power required in Btu/hr and tons for both designs"
"-------------------------------------------------------------------------------------------------------------------------------
----------"
q_steady_state = q_conv + q_rad
"-------------------------------------------------------------------------------------------------------------------------------
----------"
"END"
"-------------------------------------------------------------------------------------------------------------------------------
----------"
```

Appendix B - Results from the first EES script

8-hours case

$A = 450$	$\alpha = 0.0001735$	$alpha2 = 0.0002072$	$\alpha_{abs} = 0.39$ [-]
$\varepsilon = 0.87$	$\varepsilon_{ice} = 0.95$	$\varepsilon_{tent} = 0.87$	$F_{ice,surr} = 0.5$
$h = 8.715$	$Height = 8$ [ft]	$HF = 333.6$	$h_{tent} = 288.2$
$m_{ice} = 34358$	$\nu = 0.0001252$	$nu2 = 0.0001481$	$\nu_{L,c} = 1585$
$q_{freeze} = 1.433E+06$	$q_{ice,cool} = 143728$	$q_{net} = 673477$	$q_{rad} = 673477$
$q_{steady,state} = 893445$	$q_{surr,ice} = 191911$	$q_{tent} = 1.285E+06$	$\rho_{ice} = 916.2$
$thickness = 0.08333$ [ft]	$T_{air} = 80$ [F]	$T_{air,R} = 539.8$	$T_{film} = 52$
$T_{ice,R} = 483.7$	$T_{initial} = 52$ [F]	$T_{initial,R} = 511.7$	$T_{sky} = 55$ [F]

$B = 0.001954$	$B2 = 0.001954$	$cp_{ice} = 4.183$	$cp_{water} = 4.18$
$F_{tent,ice} = 0.5$	$F_{tent,surr} = 0.5$	$g = 4.173E+08$ [ft/hr^2]	$Gs = 291.6$ [Btu/hr*ft^2]
$k = 0.0275$	$k2 = 0.02988$	$L = 30$ [ft]	$L_c = 5$
$\nu_{L,c,2} = 48226$	$P_{atm} = 14.7$ [psia]	$q_{conv} = 219968$	$q_{cool,water} = 359012$
$q_{rad,exch} = 142240$	$q_{rad,ice} = 422721$	$q_{rad,surr} = 250756$	$q_{startup} = 2.829E+06$
$R_{a,L,c} = 2.627E+17$	$R_{a,L,c,2} = 3.323E+16$	$\sigma = 5.670E-08$ [W/m^2*K^4]	$t = 8$ [hr]
$T_{film2} = 85$	$T_{final} = 32$ [F]	$T_{final,R} = 491.7$	$T_{ice} = 24$ [F]
$T_{sky,R} = 514.7$	$T_{tent} = 90$ [F]	$T_{tent,R} = 549.7$	$W = 15$ [ft]

24-hours case

$A = 450$	$\alpha = 0.0001735$	$alpha2 = 0.0002072$	$\alpha_{abs} = 0.39$ [-]
$\varepsilon = 0.87$	$\varepsilon_{ice} = 0.95$	$\varepsilon_{tent} = 0.87$	$F_{ice,surr} = 0.5$
$h = 8.715$	$Height = 8$ [ft]	$HF = 333.6$	$h_{tent} = 288.2$
$m_{ice} = 34358$	$\nu = 0.0001252$	$nu2 = 0.0001481$	$\nu_{L,c} = 1585$
$q_{freeze} = 477578$	$q_{ice,cool} = 47909$	$q_{net} = 673477$	$q_{rad} = 673477$
$q_{steady,state} = 893445$	$q_{surr,ice} = 191911$	$q_{tent} = 1.285E+06$	$\rho_{ice} = 916.2$
$thickness = 0.08333$ [ft]	$T_{air} = 80$ [F]	$T_{air,R} = 539.8$	$T_{film} = 52$
$T_{ice,R} = 483.7$	$T_{initial} = 52$ [F]	$T_{initial,R} = 511.7$	$T_{sky} = 55$ [F]

$B = 0.001954$	$B2 = 0.001954$	$cp_{ice} = 4.183$	$cp_{water} = 4.18$
$F_{tent,ice} = 0.5$	$F_{tent,surr} = 0.5$	$g = 4.173E+08$ [ft/hr^2]	$Gs = 291.6$ [Btu/hr*ft^2]
$k = 0.0275$	$k2 = 0.02988$	$L = 30$ [ft]	$L_c = 5$
$\nu_{L,c,2} = 48226$	$P_{atm} = 14.7$ [psia]	$q_{conv} = 219968$	$q_{cool,water} = 119671$
$q_{rad,exch} = 142240$	$q_{rad,ice} = 422721$	$q_{rad,surr} = 250756$	$q_{startup} = 1.539E+06$
$R_{a,L,c} = 2.627E+17$	$R_{a,L,c,2} = 3.323E+16$	$\sigma = 5.670E-08$ [W/m^2*K^4]	$t = 24$ [hr]
$T_{film2} = 85$	$T_{final} = 32$ [F]	$T_{final,R} = 491.7$	$T_{ice} = 24$ [F]
$T_{sky,R} = 514.7$	$T_{tent} = 90$ [F]	$T_{tent,R} = 549.7$	$W = 15$ [ft]

Appendix C - Second EES Script

```
"-------------------------------------------------------------------------------------------------------------------------
----------"
" Project: Portable Ice Rink Design"
" File 2: Heat Transfer in Pipes, Pressure Drop Calculations, and Pump Selection"
"-------------------------------------------------------------------------------------------------------------------------
----------"
"1. Definitions and Given Data"
"-------------------------------------------------------------------------------------------------------------------------
----------"
"Geometric properties"
L = 35 [ft]        "Length of ice rink"
W = 15 [ft]        "Width of ice rink"
A = L * W          "Area of ice rink"
S = 4 [in] * convert(in, ft)        "Tube spacing"
Num_Tubes = L / S     "Number of tubes (should be 105)"
"PEX tubing dimensions"
D_i = 0.5 [in] * convert(in, ft)  "Inner diameter of PEX tubes"
D_o = 0.625 [in] * convert(in, ft) "Outer diameter of PEX tubes"
k_tube = 0.24 [Btu/hr*ft*F]      "Thermal conductivity of PEX"
"Thermal resistances"
R_ins = 0.2021 [hr*ft^2*F/Btu]           "Thermal resistance of ice and insulation"
"Temperatures"
T_s = 24 [F]     "Surface temperature"
"Calculate T_pipe based on heat transfer"
q_total = 39594 [Btu/hr]   "Total steady-state power from File 1"
R_total_surface = R_ins         "Total thermal resistance from pipe to surface"
T_pipe = T_s - (q_total * R_total_surface) / A   "Calculating T_pipe indirectly"
"Brine properties"
C = 25 [%]       "Brine concentration (CaCl2 solution)"
T_freeze = freezingpt(CaCl2, C=C)   "Freezing temperature of brine solution"
T_mi = T_freeze - 5 [F]   "Inlet temperature of brine (5°F below freezing point)"
"Mean temperature and outlet temperature"
T_mo =-20[F]   "Outlet temperature (to be determined by EES)"
T_m = (T_mi + T_mo) / 2       "Mean temperature"
"Brine properties at mean temperature"
rho_brine = density(CaCl2, T=T_m, C=C)     "Density of brine at T_m"
mu_brine = viscosity(CaCl2, T=T_m, C=C)              "Dynamic viscosity of brine at T_m"
nu_brine = mu_brine / rho_brine  "Kinematic viscosity"
k_brine = conductivity(CaCl2, T=T_m, C=C)            "Thermal conductivity of brine at T_m"
cp_brine = cp(CaCl2, T=T_m, C=C)  "Specific heat capacity of brine at T_m"
Pr_brine = prandtl(CaCl2, T=T_m, C=C)   "Prandtl number"
"Flow parameters"
```

```
Num_Loops = 15        "Adjusted number of loops to get integer tubes per loop"
Tubes_per_Loop = Num_Tubes / Num_Loops  "Number of tubes per loop (105/15 = 7)"
x = Tubes_per_Loop * W   "Length of tubing per loop"
v_brine = 3 [ft/s]         "Assumed velocity of brine in tubes"
A_tube = pi * D_i^2 / 4
"Variables for equation count"
Re_D = (v_brine * D_i) / nu_brine
f = moodychart(Re_D, 0)   "Assuming smooth pipe"
f8 = f / 8
Nu_D = (f8 * (Re_D - 1000) * Pr_brine) / (1 + 12.7 * sqrt(f8) * (Pr_brine^(2/3) - 1))
h = Nu_D * k_brine / D_i
R_total = (1 / (h * A_tube)) + (ln(D_o / D_i) / (2 * pi * k_tube))
U = 1 / R_total
"Mass flow rate per loop (lbm/s)"
m_dot_loop = rho_brine * A_tube * v_brine
"Convert mass flow rate to lbm/hr"
m_dot_loop_hr = m_dot_loop * 3600 [s/hr]
"Total mass flow rate"
m_dot_total = Num_Loops * m_dot_loop
m_dot_total_hr = m_dot_total * 3600 [s/hr]
"Temperature drop calculations"
Theta = exp( - U * x / (m_dot_loop * cp_brine) )
"T_mo = T_pipe - (T_pipe - T_mi) * Theta"
DeltaT_loop = T_mo - T_mi
"Heat transfer calculation"
q_totalcalculated = m_dot_total * cp_brine * DeltaT_loop
"Commented out the redundant equation"
"q_totalcalculated = q_total"   "Ensure this equation is satisfied"
"Pressure drop calculations"
"Convert rho_brine to slug/ft^3"
rho_brine_slug = rho_brine / 32.174 [lbm/slug]
deltaP_friction = f * (x / D_i) * (rho_brine_slug * v_brine^2 / 2)
Num_Bends = Tubes_per_Loop - 1
K_bend = 0.2   "Assumed loss coefficient per 180-degree bend"
deltaP_bends = Num_Bends * K_bend * (rho_brine_slug * v_brine^2 / 2)
deltaP_loop = deltaP_friction + deltaP_bends
"Pressure drop in headers (PVC pipes)"
D_header = 1.5 [in]*convert(in, ft)
A_header = pi * D_header^2 / 4
"Convert m_dot_total to lbm/s for consistent units"
m_dot_total_s = m_dot_total_hr / 3600 [s/hr]
v_header = m_dot_total_s / (rho_brine * A_header)
Re_header = (v_header * D_header) / nu_brine
f_header = moodychart(Re_header, 0)
```

```
L_header = 30 [ft]  "Adjusted length per instruction"
deltaP_header = f_header * (L_header / D_header) * (rho_brine_slug * v_header^2 / 2)
"Pressure drop in hoses"
D_hose = 2 [in]*convert(in, ft)
A_hose = pi * D_hose^2 / 4
v_hose = m_dot_total_s / (rho_brine * A_hose)
Re_hose = (v_hose * D_hose) / nu_brine
f_hose = moodychart(Re_hose, 0)
L_hose = 70 [ft]  "Adjusted length per instruction"
deltaP_hose = f_hose * (L_hose / D_hose) * (rho_brine_slug * v_hose^2 / 2)
"Pressure drop through heat exchanger"
deltaP_HX = 5 [psi]*144 [in^2/ft^2]   "Convert psi to lb/ft^2"
"Total system pressure drop"
deltaP_total = deltaP_loop + deltaP_header + deltaP_hose + deltaP_HX
"Convert deltaP_total to feet of fluid"
g_c = 32.174 [lbm*ft/(lbf*s^2)]  "Unit conversion factor"
Pump_head = deltaP_total / (rho_brine * g_c)
"Calculate required pump power"
"Volumetric flow rate (ft^3/s)"
Qv_total = m_dot_total_s / rho_brine
"Pump power (ft*lbf/s)"
Pump_power = (deltaP_total * Qv_total) / Pump_efficiency
"Convert pump power to horsepower"
Pump_power_hp = Pump_power * convert(ft*lbf/s, hp)
"Assumed pump efficiency (to be updated after pump selection)"
Pump_efficiency = 0.7  "Assumed pump efficiency"
"-------------------------------------------------------------------------------------------------------------------
----------"
"END"
"-------------------------------------------------------------------------------------------------------------------
----------"
```

Appendix D - Results from the second EES script

8-hours case

$A = 525$	$A_{header} = 0.01227$	$A_{hose} = 0.02182$	$A_{tube} = 0.001364$	$C = 25$ [%]
$\delta P_{hose} = 1305$	$\delta P_{HX} = 720$	$\delta P_{loop} = 12462$	$\delta P_{total} = 16701$	$\delta T_{loop} = 14.05$
$f = 0.02789$	$f8 = 0.003486$	$f_{header} = 0.01905$	$f_{hose} = 0.02029$	$g_c = 32.17$ [lbm*ft/(lbf*s^2)]
$L = 35$ [ft]	$L_{header} = 30$ [ft]	$L_{hose} = 70$ [ft]	$\mu_{brine} = 0.0105$	$\dot{m}_{loop} = 5.098$
$Num_{Bends} = 6$	$Num_{Loops} = 15$	$Num_{Tubes} = 105$	$\nu_{brine} = 0.000008426$	$\nu_D = 243.9$
$Pump_{power,hp} = 2.662$	$Q\nu_{total} = 0.06136$	$q_{total} = 69577$ [Btu/hr]	$q_{totalcalculated} = 2983$	$Re_D = 14835$
$R_{ins} = 0.2021$ [hr*ft^2*F/Btu]	$R_{total} = 0.3981$	$R_{total,surface} = 0.2021$	$S = 0.3333$ [ft]	$\theta = 8.125E-09$
$T_{mo} = -20$ [F]	$T_{pipe} = -2.784$	$T_s = 24$ [F]	$U = 2.512$	$v_{brine} = 3$ [ft/s]

$cp_{brine} = 2.777$	$\delta P_{bends} = 209.2$	$\delta P_{friction} = 12253$	$\delta P_{header} = 2213$
$D_{header} = 0.125$ [ft]	$D_{hose} = 0.1667$ [ft]	$D_i = 0.04167$ [ft]	$D_o = 0.05208$ [ft]
$h = 2932$	$K_{bend} = 0.2$	$k_{brine} = 0.5009$	$k_{tube} = 0.24$ [Btu/hr*ft*F]
$\dot{m}_{loop,hr} = 18354$	$\dot{m}_{total} = 76.48$	$\dot{m}_{total,hr} = 275316$	$\dot{m}_{total,s} = 76.48$
$Pr_{brine} = 58.22$	$Pump_{efficiency} = 0.7$	$Pump_{head} = 0.4165$	$Pump_{power} = 1464$
$Re_{header} = 74173$	$Re_{hose} = 55630$	$\rho_{brine} = 1246$	$\rho_{brine,slug} = 38.74$
$Tubes_{per,Loop} = 7$	$T_{freeze} = -29.05$	$T_m = -27.02$	$T_{mi} = -34.05$
$v_{header} = 5$	$v_{hose} = 2.813$	$W = 15$ [ft]	$x = 105$

24-hours case

$A = 525$	$A_{header} = 0.01227$	$A_{hose} = 0.02182$	$A_{tube} = 0.001364$	$C = 25$ [%]
$\delta P_{hose} = 1305$	$\delta P_{HX} = 720$	$\delta P_{loop} = 12462$	$\delta P_{total} = 16701$	$\delta T_{loop} = 14.05$
$f = 0.02789$	$f8 = 0.003486$	$f_{header} = 0.01905$	$f_{hose} = 0.02029$	$g_c = 32.17$ [lbm*ft/(lbf*s^2)]
$L = 35$ [ft]	$L_{header} = 30$ [ft]	$L_{hose} = 70$ [ft]	$\mu_{brine} = 0.0105$	$\dot{m}_{loop} = 5.098$
$Num_{Bends} = 6$	$Num_{Loops} = 15$	$Num_{Tubes} = 105$	$\nu_{brine} = 0.000008426$	$\nu_D = 243.9$
$Pump_{power,hp} = 2.662$	$Q\nu_{total} = 0.06136$	$q_{total} = 39594$ [Btu/hr]	$q_{totalcalculated} = 2983$	$Re_D = 14835$
$R_{ins} = 0.2021$ [hr*ft^2*F/Btu]	$R_{total} = 0.3981$	$R_{total,surface} = 0.2021$	$S = 0.3333$ [ft]	$\theta = 8.125E-09$
$T_{mo} = -20$ [F]	$T_{pipe} = 8.758$	$T_s = 24$ [F]	$U = 2.512$	$v_{brine} = 3$ [ft/s]

$cp_{brine} = 2.777$	$\delta P_{bends} = 209.2$	$\delta P_{friction} = 12253$	$\delta P_{header} = 2213$
$D_{header} = 0.125$ [ft]	$D_{hose} = 0.1667$ [ft]	$D_i = 0.04167$ [ft]	$D_o = 0.05208$ [ft]
$h = 2932$	$K_{bend} = 0.2$	$k_{brine} = 0.5009$	$k_{tube} = 0.24$ [Btu/hr*ft*F]
$\dot{m}_{loop,hr} = 18354$	$\dot{m}_{total} = 76.48$	$\dot{m}_{total,hr} = 275316$	$\dot{m}_{total,s} = 76.48$
$Pr_{brine} = 58.22$	$Pump_{efficiency} = 0.7$	$Pump_{head} = 0.4165$	$Pump_{power} = 1464$
$Re_{header} = 74173$	$Re_{hose} = 55630$	$\rho_{brine} = 1246$	$\rho_{brine,slug} = 38.74$
$Tubes_{per,Loop} = 7$	$T_{freeze} = -29.05$	$T_m = -27.02$	$T_{mi} = -34.05$
$v_{header} = 5$	$v_{hose} = 2.813$	$W = 15$ [ft]	$x = 105$

Appendix E - Third EES Script

```
"-------------------------------------------------------------------------------------------------------------------
----------"
" Project: Portable Ice Rink Design"
" File 3: Expansion Tank Sizing and Water Chiller Selection"
"-------------------------------------------------------------------------------------------------------------------
----------"
"1. Definitions and Given Data"
"-------------------------------------------------------------------------------------------------------------------
----------"
"System specifications"
V_system = 500 [gal] * convert(gal, ft^3)  "Total brine volume in ft^3"
"Brine properties"
C = 25 [%]       "Brine concentration (CaCl2 solution)"
T_freeze = freezingpt(CaCl2, C=C)   "Freezing temperature of brine solution"
"Temperatures"
T_mi = T_freeze - 5 [F]   "Inlet temperature of brine (5°F below freezing point)"
T_mo = -15 [F]   "Outlet temperature of brine (from File 2)"  "Assign actual value"
T_m = (T_mi + T_mo) / 2   "Mean brine temperature"
T_min = T_mi   "Minimum temperature from File 2"
T_max = T_mo   "Maximum temperature from File 2"
P_fill = 15 [psig]          "System fill pressure"
P_max = 40 [psig]       "Maximum system pressure"
"Brine properties at T_min and T_max"
rho_min = density(CaCl2, T=T_max, C=C)   "Density at T_max"F
rho_max = density(CaCl2, T=T_min, C=C)   "Density at T_min"
"Calculate volumetric expansion"
DeltaV = V_system * ( (1 / rho_min) - (1 / rho_max) ) * rho_max   "Volume change due to
density change"
"Calculate expansion tank size"
P_fill_abs = P_fill + 14.7 [psi]
P_max_abs = P_max + 14.7 [psi]
V_tank_min = DeltaV / ( (P_max_abs / P_fill_abs) - 1 )
"Select a standard expansion tank size"
V_tank_selected = V_tank_min * 1.2   "Selecting tank size with 20% safety margin"
"Cooling load from File 1"
q_total = 24602 [Btu/hr]   "Total steady-state power from File 1"
q_cooling = q_total
"Convert cooling load to tons"
q_cooling_tons = q_cooling / 12000 [Btu/hr per ton]
"Select chiller capacity"
Chiller_capacity = q_cooling_tons * 1.1   "Adding 10% safety margin"
"Refer to Figure 9 for chiller selection"
```

```
"Based on brine temperature and required capacity, select appropriate chiller model"
"Calculate chiller performance parameters"
"From selected chiller data (e.g., Model 8L2 at -15°F)"
COP_chiller = 3.0  "Assumed coefficient of performance"
q_input = q_cooling / COP_chiller
"Convert input power to kW"
P_input_kW = q_input * convert(Btu/hr, kW)
"-------------------------------------------------------------------------------------------------------------------------------"
"Overall conductance calculations"
"-------------------------------------------------------------------------------------------------------------------------------"
"Using LMTD method for heat exchanger"
T_ref = T_mi - 10 [F]  "Refrigerant temperature 10°F below brine inlet"
del_T1 = T_mi - T_ref
del_T2 = T_mo - T_ref
delta_T_lm = (del_T1 - del_T2) / ln(del_T1 / del_T2)
"From File 2:"
Num_Loops = 15   "Number of loops (from File 2)"
x = 105 [ft]       "Length of tubing per loop (from File 2)"
D_i = 0.5 [in]*convert(in, ft)  "Inner diameter of PEX tubes"
"Total heat transfer area (ft^2)"
A_total = Num_Loops * x * pi * D_i
"Overall conductance (UA)"
UA = q_cooling / delta_T_lm
"Calculate U based on UA and A_total"
U = UA / A_total
"Mass flow rates"
m_dot_loop = 0.1 [lbm/s]  "Mass flow rate per loop (from File 2)"
m_dot_total = Num_Loops * m_dot_loop  "Total mass flow rate"
"Calculate cp_brine"
cp_brine = cp(CaCl2, T=T_m, C=C)  "Specific heat capacity of brine at mean temperature"
"Calculate NTU"
C_min = m_dot_total * cp_brine
NTU = UA / C_min
"Effectiveness (E)"
E = 1 - exp(-NTU)
"Verify refrigerant temperature"
T_ref_calculated = T_mo - (T_mo - T_mi) / E
"-------------------------------------------------------------------------------------------------------------------------------"
"END"
"-------------------------------------------------------------------------------------------------------------------------------"
```

Appendix F - Results from the third EES script

8-hours case

A_{total} = 206.2	C = 20 [%]	$Chiller_{capacity}$ = 2.255	$COP_{chiller}$ = 3	cp_{brine} = 2.99
D_i = 0.04167 [ft]	E = 1	$\dot{m}_{loop}$ = 5.098 [lbm/s]	$\dot{m}_{total}$ = 76.47	NTU = 12.7
$P_{max,abs}$ = 54.7	$q_{cooling}$ = 24602	$q_{cooling,tons}$ = 2.05	q_{input} = 8201	q_{total} = 24602 [Btu/hr]
T_{mi} = -23.26	T_{min} = -23.26	T_{mo} = -15 [F]	T_{ref} = -28.26	$T_{ref,calculated}$ = -23.26
x = 105 [ft]				

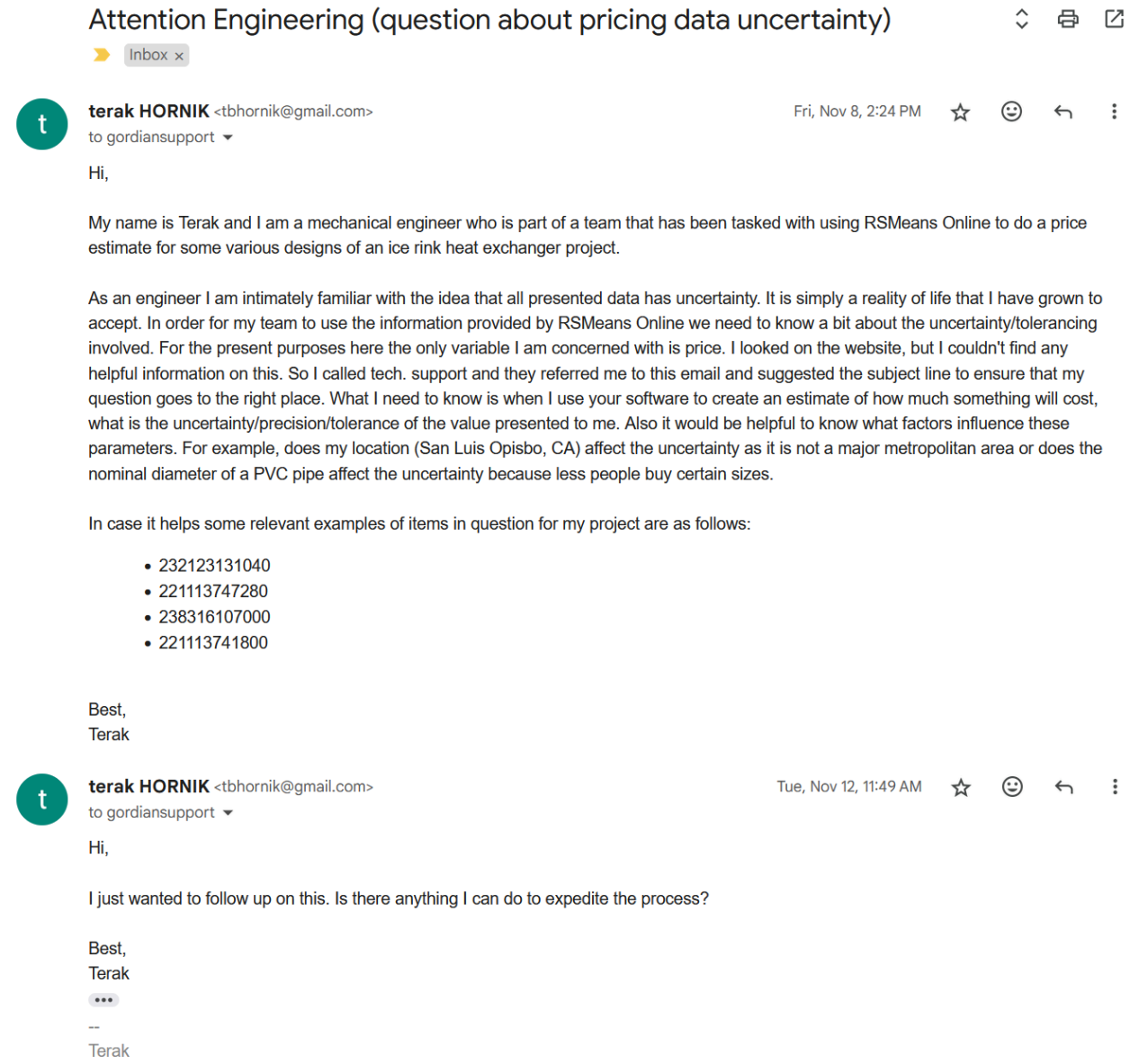
C_{min} = 228.7	δV = 0.2015	$\delta_{T,lm}$ = 8.469	del_{T1} = 5	del_{T2} = 13.26
Num_{Loops} = 15	P_{fill} = 15 [psig]	$P_{fill,abs}$ = 29.7	$P_{input,kW}$ = 2.403	P_{max} = 40 [psig]
ρ_{max} = 1192	ρ_{min} = 1188	T_{freeze} = -18.26	T_m = -19.13	T_{max} = -15
U = 14.09	UA = 2905	V_{system} = 66.84 [ft^3]	$V_{tank,min}$ = 0.2394	$V_{tank,selected}$ = 0.2873

8-hours case

A_{total} = 206.2	C = 20 [%]	$Chiller_{capacity}$ = 2.255	$COP_{chiller}$ = 3	cp_{brine} = 3.009
D_i = 0.04167 [ft]	E = 1	$\dot{m}_{loop}$ = 5.098 [lbm/s]	$\dot{m}_{total}$ = 76.47	NTU = 12.63
$P_{max,abs}$ = 54.7	$q_{cooling}$ = 24602	$q_{cooling,tons}$ = 2.05	q_{input} = 8201	q_{total} = 24602 [Btu/hr]
T_{mi} = -18.26	T_{min} = -18.26	T_{mo} = -10 [F]	T_{ref} = -23.26	$T_{ref,calculated}$ = -18.26
x = 105 [ft]				

C_{min} = 230.1	δV = 0.1549	$\delta_{T,lm}$ = 8.469	del_{T1} = 5	del_{T2} = 13.26
Num_{Loops} = 15	P_{fill} = 15 [psig]	$P_{fill,abs}$ = 29.7	$P_{input,kW}$ = 2.403	P_{max} = 40 [psig]
ρ_{max} = 1190	ρ_{min} = 1187	T_{freeze} = -18.26	T_m = -14.13	T_{max} = -10
U = 14.09	UA = 2905	V_{system} = 66.84 [ft^3]	$V_{tank,min}$ = 0.184	$V_{tank,selected}$ = 0.2209

Appendix G - Record of request for uncertainty information



Attention Engineering (question about pricing data uncertainty)

Inbox ×

terak HORNIK <tbhornik@gmail.com> Fri, Nov 8, 2:24 PM

to gordiansupport

Hi,

My name is Terak and I am a mechanical engineer who is part of a team that has been tasked with using RSMeans Online to do a price estimate for some various designs of an ice rink heat exchanger project.

As an engineer I am intimately familiar with the idea that all presented data has uncertainty. It is simply a reality of life that I have grown to accept. In order for my team to use the information provided by RSMeans Online we need to know a bit about the uncertainty/tolerancing involved. For the present purposes here the only variable I am concerned with is price. I looked on the website, but I couldn't find any helpful information on this. So I called tech. support and they referred me to this email and suggested the subject line to ensure that my question goes to the right place. What I need to know is when I use your software to create an estimate of how much something will cost, what is the uncertainty/precision/tolerance of the value presented to me. Also it would be helpful to know what factors influence these parameters. For example, does my location (San Luis Opisbo, CA) affect the uncertainty as it is not a major metropolitan area or does the nominal diameter of a PVC pipe affect the uncertainty because less people buy certain sizes.

In case it helps some relevant examples of items in question for my project are as follows:

- 232123131040
- 221113747280
- 238316107000
- 221113741800

Best,
Terak

terak HORNIK <tbhornik@gmail.com> Tue, Nov 12, 11:49 AM

to gordiansupport

Hi,

I just wanted to follow up on this. Is there anything I can do to expedite the process?

Best,
Terak

•••

--
Terak

terak HORNIK <tbhornik@gmail.com> Thu, Nov 14, 10:38 AM

to gordiansupport

Hi,

I am following up again as I have not heard back yet. When can I expect to hear back?

Best,
Terak

•••

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Terak

Support Fri, Nov 15, 9:45 AM

to me

Good Afternoon, Terak.
I consulted with our Data & Engineering team regarding your request and they provided the following information:

> "RSMeans Estimating research will contain some level of uncertainty, as you have pointed out. Our goal is to provide the costs as we've researched them, so the greatest opportunities for variance come from specification misalignment, location differences, or cost volatility over time. Unfortunately we do not have a specific percentage that can be applied since every job and estimate is different."

We hope this helps provide some context to the search results in the data. If you have any additional questions, feel free to contact us.

Sincerely,

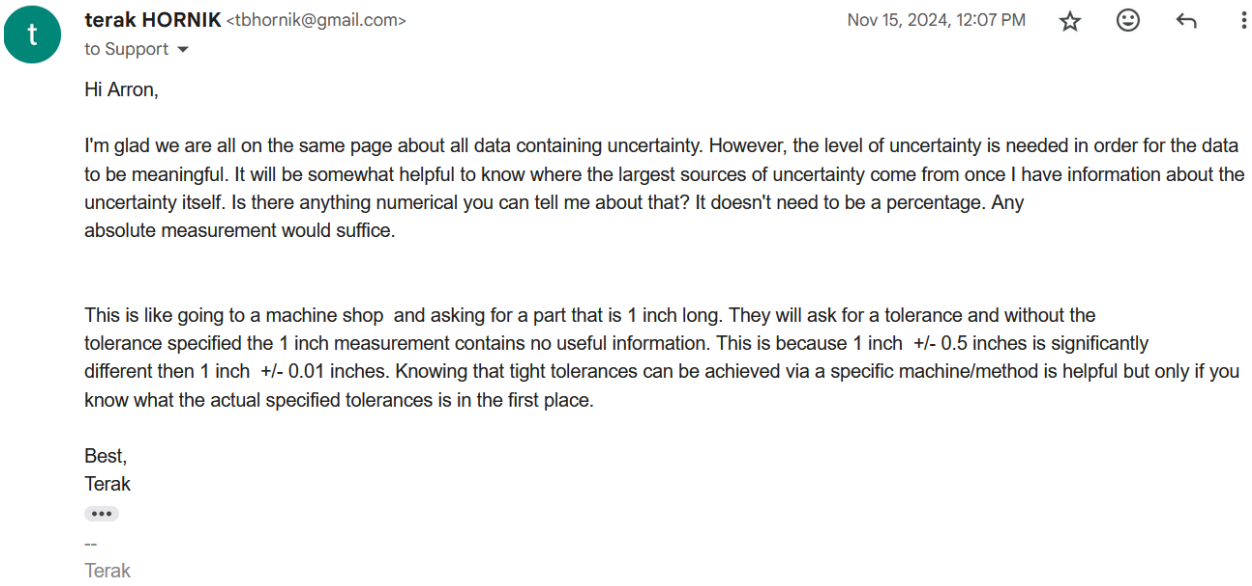
Aaron Benson
Customer Support Specialist
p: 800.448.8182
a: 30 Patewood Drive, Suite 350, Greenville, SC 29615
w: www.gordian.com | w: www.rsmeans.com

---------- Original Message ----------
From: terak HORNIK [tbhornik@gmail.com]
Sent: 11/8/2024 5:24 PM
To: gordiansupport@gordian.com
Subject: [EXTERNAL] Attention Engineering (question about pricing data uncertainty)

CAUTION: This email originated from outside of the organization. Do not click links or open attachments unless you recognize the sender and know the content is safe.

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ref:!00D400N3jP.!500VT0FpqP2:ref



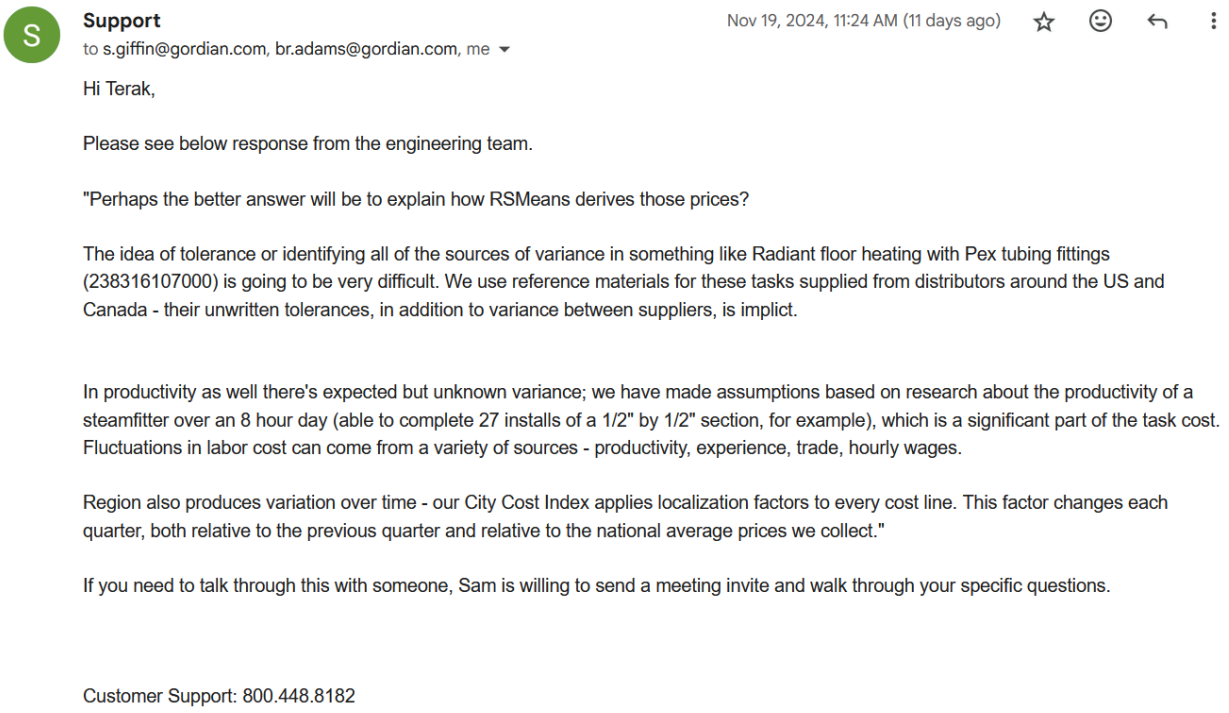
terak HORNIK <tbhornik@gmail.com> Nov 15, 2024, 12:07 PM
to Support

Hi Arron,

I'm glad we are all on the same page about all data containing uncertainty. However, the level of uncertainty is needed in order for the data to be meaningful. It will be somewhat helpful to know where the largest sources of uncertainty come from once I have information about the uncertainty itself. Is there anything numerical you can tell me about that? It doesn't need to be a percentage. Any absolute measurement would suffice.

This is like going to a machine shop and asking for a part that is 1 inch long. They will ask for a tolerance and without the tolerance specified the 1 inch measurement contains no useful information. This is because 1 inch +/- 0.5 inches is significantly different then 1 inch +/- 0.01 inches. Knowing that tight tolerances can be achieved via a specific machine/method is helpful but only if you know what the actual specified tolerances is in the first place.

Best,
Terak
•••
--
Terak



Support Nov 19, 2024, 11:24 AM (11 days ago)
to s.giffin@gordian.com, br.adams@gordian.com, me

Hi Terak,

Please see below response from the engineering team.

"Perhaps the better answer will be to explain how RSMeans derives those prices?

The idea of tolerance or identifying all of the sources of variance in something like Radiant floor heating with Pex tubing fittings (238316107000) is going to be very difficult. We use reference materials for these tasks supplied from distributors around the US and Canada - their unwritten tolerances, in addition to variance between suppliers, is implict.

In productivity as well there's expected but unknown variance; we have made assumptions based on research about the productivity of a steamfitter over an 8 hour day (able to complete 27 installs of a 1/2" by 1/2" section, for example), which is a significant part of the task cost. Fluctuations in labor cost can come from a variety of sources - productivity, experience, trade, hourly wages.

Region also produces variation over time - our City Cost Index applies localization factors to every cost line. This factor changes each quarter, both relative to the previous quarter and relative to the national average prices we collect."

If you need to talk through this with someone, Sam is willing to send a meeting invite and walk through your specific questions.

Customer Support: 800.448.8182

Some key things to note in this email thread:

- The delays in response times.
- The vague nature of the responses.
- The number of times needed to follow up to get a response.
- Recall that this was after 4 phone calls were made.
- The avoidance of the actual questions asked.
- Minimal displayed understanding of the question at hand.

Appendix H - Grading notes for Dr. Pascual

On the preliminary report there were things that we spoke to you about and you agreed that we should not lose points for them. One was significant figures. With your permission we have elected to present values as given by EES/ABAQUS/Calculators with whatever number of digits the various calculation methods result in. Other examples (that we got points back on but ‘fixed’ anyways) include the capitalization of ‘figure’ and table headings on top of the tables. We figured it was worth reminding you of this when you read our report in case you forgot.

Also we want to ask that you keep in mind how we lost a huge amount of work that EES deleted (1 large instance of lost work with many many more less significant ones) and thus the work shown here is not 100% of what we actually did.

Lastly, we want to say that we know we are unlikely to be 100% correct in all of our values presented here. In an ideal world we would have more time to debug the EES files (well actually, in a really ideal world EES wouldn't exist…) and do a better job of checking all the values and tracing any issues we came across. However, it was a huge triumph that we got the codes working in the first place and while we know that the answers are in all likelihood not all correct we feel we put a reasonable amount of time into this for a lab project. Also FYI we did an estimated 19 hours of debugging (without 100% success) after you sent us feedback and gave an estimate of 3 hours of work on EES remaining. As engineers we feel it is critically important to include our own level of uncertainty in the displayed results, so we have done so here.